

A Frequency-Sensitive Competitive Learning Networks with Hadamard Transform Applied to Color Image Compression

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Abstract

The neural network is useful for data compression if the connection weights are chosen properly. In this paper, a Modified Frequency-Sensitive Competitive Learning (MFSCCL) network with Hadamard transform based on Vector Quantization (VQ) for color image compression is presented. The goal is apply a spread-unsupervised scheme based on the modified competitive learning networks so that on-line learning and parallel implementation for color image compression based on VQ in Hadamard Transform (HT) domain are feasible. In the MFSCCL network, each output neuron (codevector) incorporates a count of the number of times it has been the winner. The distortion measure used to determine the winner is updated to include the count number. The color image information transformed by HT operation was separated into RGB 3-plane DC value and AC coefficients. Then the AC coefficients for each plane were trained using the proposed MFSCCL method to generate the VQ codebook. The experimental results show that promising codebooks can be obtained using the presented spread MFSCCL network for color image compression in the transform domains.

Keywords: data compression, neural networks, Hadamard transform.

1 Introduction

A vector quantizer is a mapping of an input or test vector to an output or reproduction vector that is selected from a finite codebook of stored vectors. A number of vector quantization algorithms for data compression have proposed over the years [1-3]. The purpose of vector quantization is to create a codebook such that the average distortion between training vectors and their corresponding codevectors in the codebook is minimized. Codebook design can be considered as a clustering process in which each training vector is classified to a specific class. The clustering process updates the codebook iteratively such that the average distortion between training vectors and codevectors in the codebook becomes smaller and smaller.

Neural network are well suited to the problem of image compression due to their massively parallel and distributed architecture. Neural networks with competitive learning have been demonstrated capable of performing vector quantization [4-6]. A problem with the training procedure of competitive learning networks is that it sometimes leads to codevectors which are not distributed in an equiprobable manner [7]. The Frequency-Sensitive Competitive Learning (FSCL) [8] network is to overcome the limitations of the conventional competitive learning network. In Modified Frequency-Sensitive Competitive Learning

(MFSCl) network, the learning rule and stopping criterion of the FSCL network are modified to address the codebook design. The problem of VQ is regarded as a minimization process of an object function. This object function is defined as the average distortion between the training vectors in a divided image to the cluster centers represented by the codevectors in the codebook. The modified frequency-sensitive competitive learning network is simpler than that of the FSCL network, and is constructed as a two-layer fully interconnected array with the input neurons representing the training vectors and output neurons representing the codevectors in the codebook.

Color images comprise three planes: red, green, and blue. Thus, the most straightforward method to encode a color image by modified frequency-sensitive competitive learning networks is to split the RGB color image into 3 planes and compress them separately by treating each color component as a single gray-level image [9]. In this paper, the MFSCl network has been extended to Spread MFSCl network and applied to color image compression.

In the Hadamard Transform (HT) domain, this object function is defined as the average distortion between the training vectors in AC coefficients to the cluster centers represented by the codevectors in the codebook. The training vectors, constructed by AC coefficients, are directly fed into the MFSCl network. The MFSCl and spread MFSCl networks are demonstrated to have the capability of VQ in the transform domains.

The rest of this paper is organized as follows. Section 2 reviews the competitive learning networks. Section 3 presents the spread MFSCl network for color VQ. Section 4 presents the Hadamard transform and spread MFSCl network. Experimental

results are given in Section 5. Finally, conclusions are drawn in Section 6.

2 Competitive Learning Network

A competitive learning network is an unsupervised network which selects a winner based on similarity measure over the feature space. A proper neuron state is updated if and only if it wins the competition among all neurons. Many schemes for competitive learning networks have been proposed [10-12].

In the simple competitive learning network, the single output layer consists of cluster centers, each of which is fully connected to the inputs via interconnection strength. In the conventional competitive learning only one output unit is active at a time and the objective function is given by

$$J_c = \frac{1}{2} \sum_{j=1}^c \sum_{i=1}^n u_{i,j} \|\mathbf{x}_i - \mathbf{w}_j\|^2 \quad (1)$$

where n and c are the number of training vectors and the number of clusters respectively. $u_{i,j} = 1$ if $\mathbf{x}_i$ belongs to cluster j and $u_{i,j} = 0$ for all other clusters. The neuron that wins the competition is called the winner-take-all neuron. Thus $u_{i,j}$ indicates whether the input sample $\mathbf{x}_i$ activates neuron j to be a winner. $u_{i,j}$ is given by

$$u_{i,j} = \begin{cases} 1 & \text{if } \|\mathbf{x}_i - \mathbf{w}_j\| \leq \|\mathbf{x}_i - \mathbf{w}_k\|, \text{ for all } k; \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

The incremental $\Delta \mathbf{w}_j$ is given by [10]

$$\langle \Delta \mathbf{w}_j \rangle = -\eta \frac{\partial J_c}{\partial \mathbf{w}_j} = \eta \sum_{i=1}^n (\mathbf{x}_i - \mathbf{w}_j) u_{i,j} \quad (3.a)$$

where $j = 1, 2, \dots, c$. Although Eq. (3.a) is written as a sum over all samples, practically it is usually used incrementally, i.e.

$$\Delta \mathbf{w}_j = \eta (\mathbf{x}_i - \mathbf{w}_j) \mu_{i,j}, \quad j = 1, 2, \dots, c. \quad (3.b)$$

The updating rule is given by

$$\mathbf{w}_j(t+1) = \mathbf{w}_j(t) + \Delta \mathbf{w}_j(t). \quad (4)$$

In the above equation, the t is the training time index, η is the learning-rate parameter, and is typically reduced monotonically to zero as learning progresses. The simple competitive learning algorithm can be described as follows.

Step 1: Apply an input sample $\mathbf{x}_i$.

Step 2: Find the distortion for all output neural units.

Step 3: Select the output unit with the smallest distortion and label it as the winner and its weight vector as $\mathbf{w}_j$

Step 4: Update the selected weight vector by Eqs. (3) and (4).

Step 5: Repeat Steps 1 through 4 for all training samples.

3 Spread MFSCl network for Color VQ

Suppose an image is divided into n blocks (vectors of pixels) and each block consists of $\ell \times \ell$ pixels. A vector quantizer is a technique that maps the Euclidean $\ell \times \ell$ -dimensional space $\mathbf{R}^{\ell \times \ell}$ into a set $\{\mathbf{w}_j, j = 1, 2, \dots, c\}$ of points in $\mathbf{R}^{\ell \times \ell}$, called a codebook. It looks for a codebook such that each training vector is approximated as close as possible by one of the code vectors in the

codebook. A codebook is optimal if the average distortion is at the minimum value. The average distortion $E[d(\mathbf{x}_i, \mathbf{w}_j)]$ between an input sequence of training vectors $\{\mathbf{x}_i, i = 1, 2, \dots, n\}$ and its corresponding output sequence of code vectors $\{\mathbf{w}_j, j = 1, 2, \dots, c\}$ is defined as

$$D = E[d(\mathbf{x}_i, \mathbf{w}_j)] = \frac{1}{n} \sum_{i=1}^n d(\mathbf{x}_i, \mathbf{w}_j) \quad (5)$$

A modified distortion measure for the training process is defined as

$$d^*(\mathbf{x}_i, \mathbf{w}_j) = d(\mathbf{x}_i, \mathbf{w}_j(t)) \times c_j(t) \quad (6)$$

where $c_j(t)$ is the total number of times that code vectors $\mathbf{w}_j$ has been the winner during training. The winning code vector at each iteration of the training process is the vector with the minimum d^* .

The MFSCl network has the same architecture as the FSCL network. It is an unsupervised competitive learning network using the modified competitive learning rule and stopping criterion. Similar to the standard competitive learning rule in Eqs. (3.b) and (4), the least squared error solution can be obtained by

$$\mathbf{w}_j(t+1) = \begin{cases} \mathbf{w}_j(t) + \eta (\mathbf{x}_i - \mathbf{w}_j) & \text{if } \|\mathbf{x}_i - \mathbf{w}_j\| \leq \|\mathbf{x}_i - \mathbf{w}_k\|, \text{ for all } k; \\ \mathbf{w}_j(t) & \text{otherwise} \end{cases} \quad (7)$$

The MFSCl algorithm modifies only output neurons without updating the interconnection strengths. Instead of updating the interconnection strengths using the winner-take-all scheme in the FSCL network and for the purpose of simplifying the hardware architecture, the MFSCl network only modifies

the output states (codevectors).

We then map R, G and B plane training vectors of a color image to the spread MFSCl neuron array that compresses them separately by treating each color plane as a single gray-level image. Therefore, the spread MFSCl network based vector quantizer in the p th plane for the modified competitive learning rule can be modified as

$$\omega_{j,p}(t+1) = \begin{cases} \omega_{j,p}(t) + \eta(\mathbf{x}_{i,p} - \omega_{j,p}) & \text{if } \|\mathbf{x}_{i,p} - \omega_{j,p}\| \leq \|\mathbf{x}_{i,p} - \omega_{k,p}\|, \text{ for all } k; \\ \omega_{j,p}(t) & \text{otherwise} \end{cases} \quad (8)$$

Then, the steps of codebook design each plane using the spread MFSCl network are given as follows.

- Step 1: Initialize the codevectors $\omega_{j,p}$ ($2 \leq j \leq c$), learning rate η , maximum error (ME), total error (TE), and a threshold value ε .
- Step 2: Input a training vector $\mathbf{x}_{i,p}$ and find the winner's codevector based on the minimum d^* .
- Step 3: Apply Eq. (8) to update the winner's codevector and set $TE = TE + \|\mathbf{x}_{i,p} - \omega_{j,p}\|$.
- Step 4: Repeat Step 2 and 3 for all input samples, then if $(ME - TE)/ME < \varepsilon$, go to step 5; otherwise replace ME content from TE , and go to Step 2.
- Step 5: Complete the codebook design in the p th plane ($p=1,2,3$).

4 Hadamard Transform and Spread MFSCl network

Among various popular image transforms,

Hadamard Transform (HT) [13-14] takes the least computation overhead since its basis vectors content only +1 and -1. No multiplication is necessary and only fixed point arithmetic is required for computing the transformation. The forward and inverse 2-D Hadamard transform for an $N \times N$ image can be define as

Forward:

$$F(u, v) = \frac{1}{N} \sum_{j=0}^{N-1} \sum_{k=0}^{N-1} f(j, k) (-1)^{\sum_{i=0}^{\ell-1} [b_i(u) b_i(j) + b_i(v) b_i(k)]} \quad (9)$$

Inverse:

$$f(j, k) = \frac{1}{N} \sum_{u=0}^{N-1} \sum_{v=0}^{N-1} F(u, v) (-1)^{\sum_{i=0}^{\ell-1} [b_i(u) b_i(j) + b_i(v) b_i(k)]} \quad (10)$$

where $F(u, v)$ are the Hadamard transform coefficients, $f(j, k)$ are the pixel value at (j, k) , $N = 2^\ell$ for some ℓ , and $b_i(u)$ is the i th bit of u in the binary form.

For a 2-D $N \times N$ image, the input image will be divided into nonoverlapping blocks of equal size. The Hadamard transform is applied to transform each block into DC value and AC coefficients. Then the AC coefficients are fed into the spread MFSCl algorithm for training vectors and generating better codebooks. In the decoder, indexes of codevectors are received to get AC information (codevectors). Then, the inverse Hadamard transform is used to reconstruct the image using DC value and AC coefficients.

5 Experimental Results

The codebook design is the primary problem in image compression based on vector quantization. In this paper, the images reconstructed from the LBG · MFSCl · Hadamard Transform + MFSCl (named by

HT+MFSCl) methods are compared. The training vectors were extracted from 256×256 with 8-bit gray level real images, each of which is divided into 4×4 blocks to generate 4096 non-overlapping 16-D training vectors. Two codebooks of size 128 and 256 were built by these training vectors. The resulting images were evaluated subjectively by the mean squared error (MSE) and peak signal to noise ratio (PSNR) that is defined for images of size $N \times N$ as

$$PSNR = 10 \log_{10} \frac{255^2}{MSE} \quad (11)$$

and

$$MSE = \frac{1}{N^2} \sum_{i=1}^N \sum_{j=1}^N (x_{ij} - \hat{x}_{ij})^2 \quad (12)$$

where x_{ij} and $\hat{x}_{ij}$ are the pixel gray levels from the original and reconstructed images, and 255 is the peak gray level. Fig. 1 shows the PSNR and MSE of the “Girl” images reconstructed from two codebooks of size 128 and 256 designed by the LBG and the presented MFSCl and HT+MFSCl methods. From experimental results, the reconstructed images obtained from the presented MFSCl network are superior to those obtained from the LBG algorithm, and the performance of the presented HT+MFSCl algorithm is significantly better than the MFSCl algorithm.

In the color compression simulation, the color image information transformed by Hadamard transform operation was separated into RGB planes DC value and AC coefficients. Then the AC coefficients for each plane were trained using the proposed spread MFSCl methods to generate better codebook based on VQ. To show the reconstruction performance, the resulting images were evaluated by the average PSNR among three-color planes

$$PSNR_A = \frac{PSNR_R + PSNR_G + PSNR_B}{3} \quad (13)$$

where $PSNR_R$, $PSNR_G$, and $PSNR_B$ are the PSNR for red, green, and blue planes, respectively. The PSNRs of the “Tree” and “Girl” images calculated are shown in Fig. 2, and their reconstructed images using the HT + spread MFSCl network with 256 codevectors each plane are shown in Figure 3. Experimental results show that the HT + spread MFSCl method produce promising reconstructed images quality.

6 Conclusions

In this paper, a spread unsupervised parallel approach called spread MFSCl network with Hadamard transform based on vector quantization for color image compression has been presented. Instead of updating the interconnection strengths using the winner-take-all scheme in the FSCL network, the MFSCl algorithm only modifies output neurons and omits the updating of the interconnection strengths. Experimental results show that the MFSCl network is valid for codebook design and the HT+MFSCl method produce promising reconstructed images than those reconstructed by the MFSCl method. In addition, an important advantage of using neural networks for vector quantization is that the computation can be carried out in parallel by the neural units. Our future work includes VHDL implementation of MFSCl algorithm. The target technology will be FPGA design.

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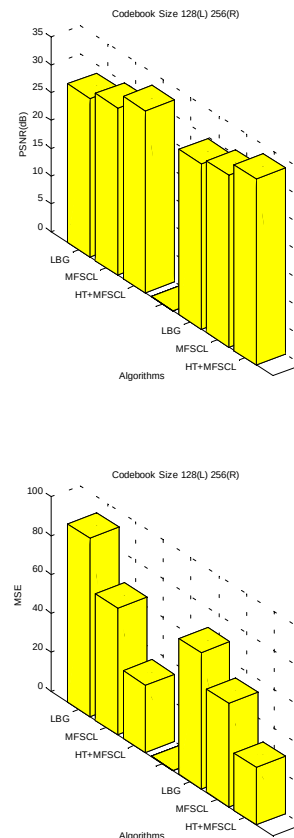


Fig. 1 PSNR and MSE of the "Girl" image reconstructed by the LBG and the presented MFSCl and HT+MFSCl algorithms.

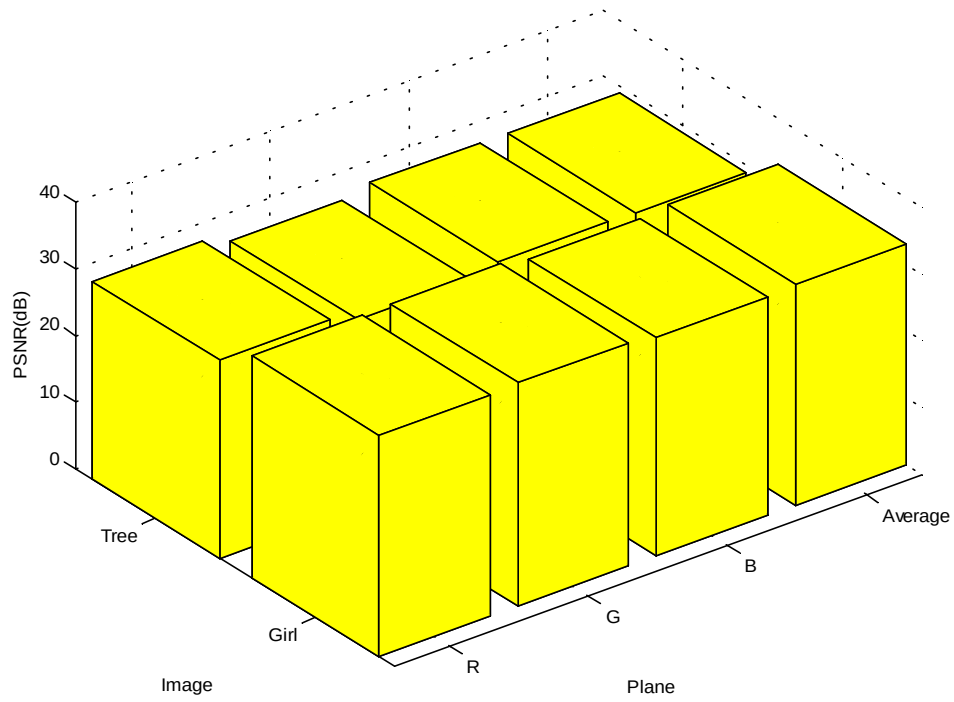


Fig. 2 PSNRs of the “Tree” and “Girl” images reconstructed by the Spread MFSCCL with 256 codevectors each plane.



Fig. 3 The “Tree” and “Girl” original color images (up row) and their reconstructed images (down row) using the HT + spread MFSCCL with 256 codevectors each plane.